KUMARAGURU COLLEGE OF TECHNOLOGY

Department of Artificial Intelligence and Data Science

DATA SCIENCE AND VISUALIZATION (24ADI204)

WEEK 7 REPORT

Team Formation & Data Hunting

Team 16: Youth Smoking and Drug Consumption Analysis

DATA-VIZ HACK-A-BIT 2026–2027

Team Member Name	Roll Number
Lakshmi Prabha M M	25BAD049

Relationship Analysis – Correlation Between Features

1. Week 7 Contribution

Analysed the correlation between numerical features to understand relationships and patterns in the dataset.

2. Work Carried Out

Studied the relationships between different numerical variables in the dataset using correlation analysis. Examined positive, negative and weak correlations to understand how features are related. Also studied how correlation results can help identify highly related variables.

3. Topics Researched / Studied

- **Correlation Analysis**
- **Positive Correlation**
- **Negative Correlation**
- **Zero or Weak Correlation**
- **Pearson Correlation Coefficient**
- **Correlation Matrix**
- **Heatmap Visualization**
- **Multicollinearity**

- **Interpretation of Feature Relationships**

4. Outcome of My Contribution

Understood how correlation analysis helps identify relationships between features. Gained knowledge of interpreting correlation values and identifying variables that may have strong relationships with each other.

5. Reference

Project notebook: DSV Team 16 – Youth Smoking and Drug Consumption Analysis.
